

S.No	Description	Theoretical	Practical
1.	R vs Pt Pt = 1:10:10000 (Pt = 9.394e dB) Gf = 30 Gf = 30 $\lambda = 0.03m$ $\sigma = 1m^2$ PR = $10^{-12} W$ Pt = 8.697 W	$R = \left[\frac{P_r G_T G_R \lambda^2 \sigma}{(4\pi)^3 P_r} \right]^{1/4}$ $R = \left[\frac{8.697 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^3 \times 10^{-12}} \right]^{1/4}$ $= \left[\frac{7.04457}{1.9844 \times 10^{-9}} \right]^{1/4}$ $= (3.5499 \times 10^9)^{1/4}$ $= 247.1m$	R = 247.1m
2.	R vs Pr Pr = -15:1 :-9/-1084 dB) Gf = 30 Gf = 30 $\lambda = 0.03m$ $\sigma = 1$ Pt = 10000 Pr = 1.445×10^{-11}	$R = \left[\frac{10000 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^3 \times 1.445 \times 10^{-11}} \right]^{1/4}$ $= \left[\frac{810}{3.9594 \times 10^{-8}} \right]^{1/4}$ $= \left[\frac{810}{3.9594 \times 10^{-8}} \right]^{1/4}$ $= 378.19$	R = 378.19

Exp 5
Date 11/9/26
Table: 7

COMPUTE PARAMETRE LIKE RANGE, VELOCITY,
POWER USING THE RADAR EQUATION.

Aim:

To compute the maximum detection range and received power of a Radar system by the use of RADAR RANGE operation.

Apparatus Required :-

- * Computer / Laptop
- * Simulink Software.

Code:

```
clc;  
clear;  
Pt = 9380;  
Gt_dB = 30;  
Gt = 10^(Gt_dB/10);  
f = 1e9;  
c = 3e8;  
lambda = c/f;  
Sigma = 100;  
L_dB = 3;
```

```
L = 10^(L_dB/10);  
L = 10^(L_dB/10);
```

```
Smin = 1e-13;
```

```
Rmax = ((Pt * Gt^2 * lambda^2 * Sigma) /  
(4 * pi^3 * Smin * L))^(1/4);
```

Calculation

$$P_t = 8.694$$

$$R = 2.472 \text{ e}^2 = 247.2$$

$$G_t, G_r = 30$$

$$R = \left[\frac{8.694 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^3 \times 10^{-12}} \right]^{1/4}$$

$$= \left(\frac{9.04457}{1.9944 \times 10^{-1}} \right)^{1/4}$$

$$= (3.5449 \times 10^9)^{1/4}$$

$$= 244.1 \text{ m}$$

$$P_R = -1.164 \text{ e}^2 = 1.99526 \times 10^{-11}$$

$$P_R = 1.99526 \times 10^{-11}$$

$$R = 3.984 \text{ e}^2$$

$$= 398.4$$

$$R = \left(\frac{810}{3.9594 \times 10^{-8}} \right)^{1/4}$$

$$R = 398.19 \text{ m}$$

```
disp (" RADAR . SIMULATION RESULT " );
```

```
disp (" Transmitted Power (W): " );
```

```
disp (Pt);
```

```
disp (" Antenna Gain (dB): " );
```

```
disp (G/10);
```

```
disp (G_dB);
```

```
disp (" Frequency (GHz): " );
```

```
disp (f/1e9);
```

```
disp (" Wavelength (m): " );
```

```
disp (sigma);
```

```
disp (" System Loss (dB): " );
```

```
disp (L_dB/10);
```

```
disp (" Minimum Detectable Signal (W): " );
```

```
disp (S_min);
```

```
disp (" Maximum Detectable Signal (W): " );
```

```
disp (" Maximum Radar Range (km): " );
```

```
disp (R_max/1000);
```

```
disp
```

```
R = 100 : 100 : 10000;
```

```
Pr = (Pt * G^2 * lambda^2 * sigma) / (4 * pi)^3 * R.^4 * L;
```

```
Pr_dBm = 10 * log10(Pr / 1e-3);
```

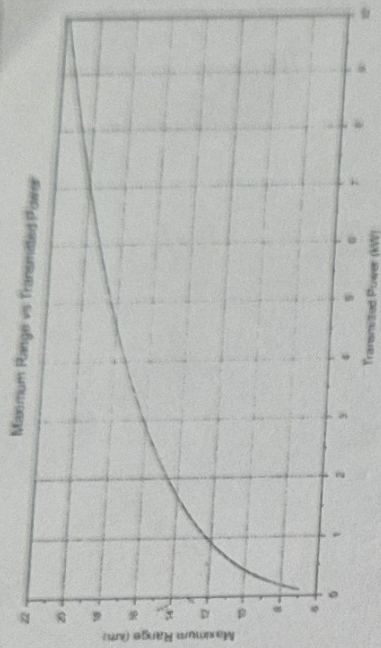
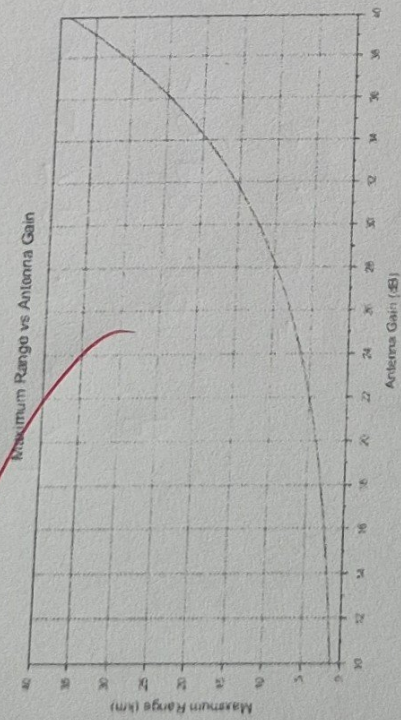
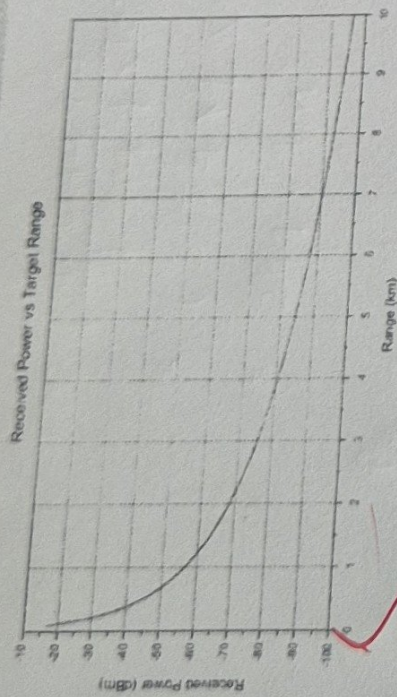
```
G_dB_range = 10 : 1 : 40;
```

```
G_range = 10.^(G_dB_range / 10);
```

```
R_max_G = ((Pt * G_range.^2 * lambda^2 * sigma) /  
(4 * pi)^3 * S_min * L)).^(1/4);
```

```
R_max_G_km = R_max_G / 1000;
```

```
set(1);
```



```

Subplot(2,2,1);
plot(R/1000, Pr_dBm);
xlabel ("Range (km)");
ylabel ("Received Power (dBm)");
title ("Received Power vs Target Range");

xgrid();

Subplot(2,2,2);
plot(Pt_range /1000, Rmax_Pt_km);
xlabel ("Transmitted Power (kW)");
ylabel ("Maximum Range (km)");
title ("Maximum Range vs Transmitted Power");

xgrid();

Subplot(2,2,3);
plot(G_dB_range, Rmax_G_km);
xlabel ("Antenna Gain (dB)");
ylabel ("Maximum Range vs Antenna Gain");

xgrid();

```

Sno	Content	Mark	Mark obtained
1.	Program implementation	20	20
2.	wave form & Tabulation	20	20
3.	Calculation and result	20	20
4.	viva - voice	20	16
5.	Result Observation	20	20
		Total	96

Result :-

Therefore the code has executed successfully. of the ~~maximum~~ detection range and received power of a Radar system by the use of Radar Range operations.